

# Prediction of Long-Term Performance and Durability of BFRP Bars under the Combined Effect of Sustained Load and Corrosive Solutions

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**Abstract:** Unlike carbon-fiber-reinforced polymer (CFRP) bars and the maturely developed glass FRP (GFRP) bars, the long-term performance of the newly developed basalt-fiber reinforced polymer (BFRP) bars under severe environmental conditions remains unclear. This paper evaluates the residual tensile properties of unstressed and stressed BFRP bars exposed to four types of simulated harsh environments: alkaline solution, salt solution, acid solution, and de-ionized water at 25, 40, and 55°C. Microstructural analysis using scanning electronic microscopy (SEM) was also conducted to reveal the inherent degradation mechanism of BFRP bars in alkaline environment. The residual tensile strength of unstressed BFRP bars exposed to alkaline solution was used for long-term performance prediction based on Arrhenius theory. The results showed that the effect on the durability of BFRP bars exposed to acid, salt, and deionized water was less than that for bars exposed to alkaline solution. The effects of sustained stress on the degradation of BFRP bars were not obvious when the stress level was less than 20% of the ultimate strength, but if the stress exceeded this level, the degradation processes were accelerated. The predicted exposure time required for a reduction in strength of 50% at a mean annual temperature of 5.7°C in alkaline solution, which represents an area with a northern latitude of 50°, was estimated at approximately 16.1 years for the 6-mm BFRP bar. DOI: 10.1061/(ASCE)CC.1943-5614.0000517. © 2014 American Society of Civil Engineers.

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## Introduction

Traditional reinforced concrete (RC) structures often deteriorate because of corrosion of the steel rebar if exposed to corrosive media (seawater, deicing salt, etc.). Various remedies (e.g., use of epoxy-coated, galvanized, or stainless steel bars) have proven to be costly and inadequate in the long run. In recent years, fiber-reinforced polymer (FRP) bars have been increasingly used in civil engineering and are expected to serve as an ideal alternative to steel bars because of their resistance to corrosion. However, because of the continuously high cost of carbon FRP (CFRP) and the relatively poor resistance of E-glass FRP to alkaline conditions, their applications as structural reinforcements are limited to a small amount or

as accessory structures. To overcome these limitations and apply the potential advantages of FRP bars to RC structures, an increasing amount of research has addressed the promising basalt-fiber-reinforced polymer (BFRP) material, which generally shows superior mechanical properties and more stable chemical resistance compared with E-glass FRP. A wider range of working temperatures (Cao et al. 2009), better resistance to the freeze-thaw cycle (Shi et al. 2013), and better fatigue performance (Wu et al. 2010) are also observed. In addition, BFRP has a much lower cost than CFRP.

Numerous studies have been conducted on the durability of FRP bars in simulated corrosive solutions (Bank et al. 2003). However, most of these studies focused on carbon, aramid, and glass FRP. Because of the short period of the development of basalt FRP and the complex environmental conditions of civil infrastructure, the degradation of BFRP bars in corrosive environments has not yet been thoroughly examined. Immersion of the FRP reinforcements in various simulated corrosive solutions and acceleration of the associated processes with elevated temperatures represent the most commonly adopted research technique found in the literature (Micelli and Nanni 2004; Chen et al. 2007; Robert et al. 2010; Kamal and Boulfiza 2011; Al-Salloum et al. 2013). Furthermore, tests on FRP bars enveloped in concrete during environmental exposure were conducted by other researchers [e.g., Robert et al. (2009) and Robert and Benmokrane (2013)]. However, these studies were all conducted without sustained loading, which is different from the actual service conditions. Indeed, the micropores in the resin matrix give rise to stress concentrations when the bars are subjected to tensile stress, which accelerates the production of microcracks. These microcracks provide additional channels for the invasion of surrounding media and have an adverse effect on the durability of FRP bars. Nkurunziza et al. (2005) found that durability tests must be performed under sustained loading to more closely resemble field conditions. Various sustained tensile loadings

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(22–68% of ultimate strength) were adopted by Benmokrane et al. (2002) in a durability investigation of GFRP bars. The GFRP bars with diameters of 15.9 mm (no. 5), 12.7 mm (no. 4), and 9.5 mm (no. 3) were aged in alkaline and water environments and subjected to sustained loading and elevated temperature in work by Debaiky et al. (2006) using a specially designed loading frame. The current design guidelines and codes—ACI440.1R-06 (ACI 2006), JSCE (1997), CAN/CSA-S806-02 (CSA 2002), and CAN/CSA-S6-00 (CSA 2000)—specify different strength reduction factors to account for the effects of sustained stress, fatigue, and environmental conditions. However, no reduction factors were currently available for BFRP bars because of a lack of sufficient research data.

Four main types of corrosive environments are often encountered in civil engineering: alkaline environments (concrete-pore solutions), salt environments containing  $\text{Cl}^-$  (seawater, deicing salt), acid environments (acid rain), and water environments (damp environments). An accelerated aging test was conducted on BFRP bars with the four types of simulated corrosive solutions combined with sustained loading levels. The effects of temperature, stress level, and corrosive solution type on the durability performance of BFRP bars were studied. With the help of the SEM images, microstructural studies were conducted. Using Arrhenius theory, a long-term prediction process was carried out for unstressed BFRP bars in alkaline solution. This paper clarifies the long-term durability properties of the newly developed innovative BFRP bars and lays a solid foundation for improving the durability of critical infrastructure constructed with them.

## Experimental Program

### Parameters Included

Three parameters were investigated in this study:

1. Temperature: Because it is known that temperature has a deep influence on the corrosion rate of FRP materials, three temperatures were selected—25, 40, and 55°C—using a temperature-controlled cabinet. Based on the degradation data for the three temperatures, a long-term performance prediction was carried out.
2. Surrounding solution: Four types of solution were used to simulate the common state of the reinforcing bars in the field: (1) deionized water, which simulates a highly humid environment; (2) alkaline solution, which simulates a cement paste; (3) salt solution, which simulates an ocean environment or the use of deicing salt; and (4) acid solution, which simulates acid rain.
3. Sustained load: In field structures, tensile stress in the FRP bars is caused by the external load. The presence of the sustained stress promotes the ability of external media to migrate through the microcracks in the polymer matrix around the fibers, thus increasing the attack on the basalt fibers. However, experimental studies on the coupling effect of stress are still lacking. In this study, an axial tensile stress was applied to the BFRP bars to more closely simulate the field environment. The stress level was maintained at 10–60% of the average ultimate tensile stress, corresponding to tensile strains of 3,000–18,000  $\mu\epsilon$ .

### Materials and Specimens

The basalt fibers were provided by Zhejiang GBF Basalt Fiber, Hengdian, China [and are shown in Fig. 1(a)]; the chosen resin was a Bisphenol-A-type epoxy resin (Blue star New Chemical Materials, Wuxi, China). The properties of the basalt fiber and

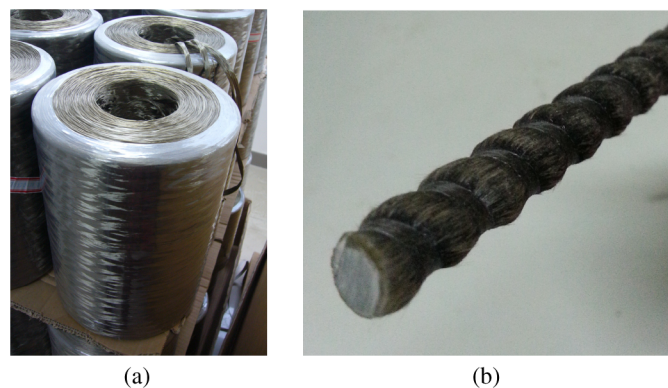


Fig. 1. Materials: (a) basalt fiber; (b) BFRP bar (images by Gang Wu)

Table 1. Properties of the Basalt Fiber and the Epoxy Resin Provided by the Manufacturers

| Item                                               | Basalt fiber | Item                                               | Epoxy resin |
|----------------------------------------------------|--------------|----------------------------------------------------|-------------|
| Density ( $\text{g}/\text{cm}^3$ )                 | 2.65         | Density ( $\text{g}/\text{cm}^3$ )                 | 1.13        |
| Tensile strength (MPa)                             | 3,800–4,840  | Impact strength (KJ/m)                             | 1.69        |
| Elastic modulus (GPa)                              | 93.1–110     | Fracture toughness (N/m)                           | 142         |
| Elongation at break (%)                            | 3.0          | Elastic modulus (GPa)                              | 3.6         |
| Maximum service temperature ( $^{\circ}\text{C}$ ) | 650          | Heat distortion temperature ( $^{\circ}\text{C}$ ) | 95          |

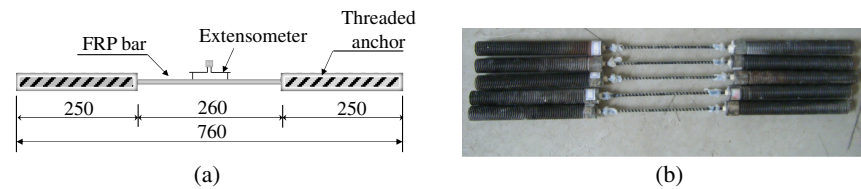
Note: The tensile strength of basalt fiber is the strength of the virgin filament.

the epoxy resin provided by the manufacturers are listed in Table 1. As shown in Fig. 1(b), BFRP bars impregnated with resin were produced using pultrusion technology. The surface of the BFRP bars was ribbed with a nylon laminate during pultrusion, and the epoxy-based BFRP bar was postcured in a 200°C oven for two hours. The fiber volume fraction was approximately 65%, and the average cross-sectional area of the bar in immersion testing was 28.52  $\text{mm}^2$ . The adopted nominal diameter of the bar was 6 mm (bar area of 28.26  $\text{mm}^2$ ) in the calculation of tensile stress.

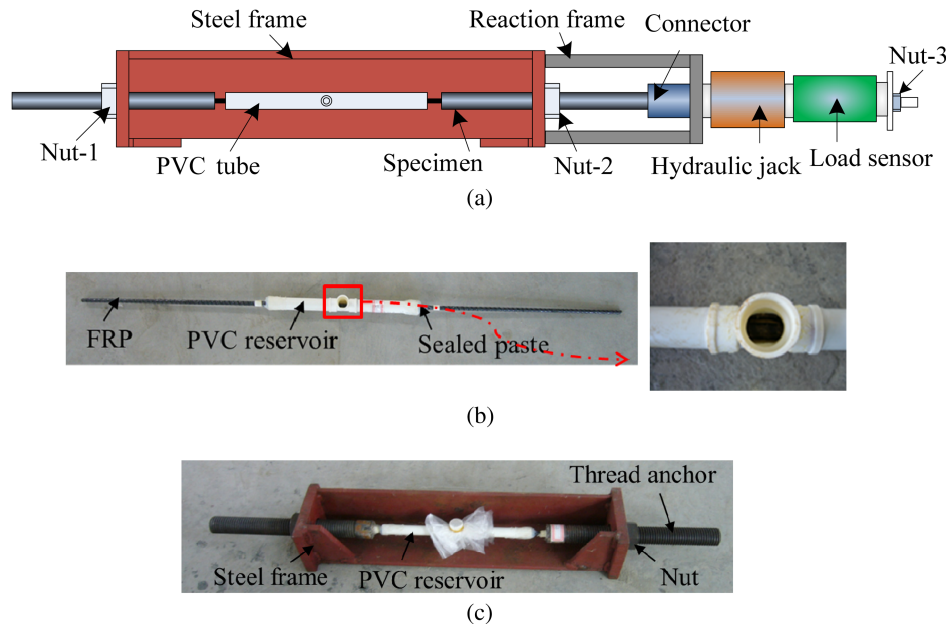
The tensile properties were tested in accordance with ACI440.3R-04 B2, and the overall length of each specimen was 760 mm. Each end was sandblasted and placed in a 250-mm-long threaded steel tube grouted with an epoxy matrix to create grip ends for axial tension. As shown in Fig. 2(a), the length of the test section was 260 mm, which is longer than 40 times the nominal diameter of the BFRP bar. The elastic modulus was obtained from the attached extensometer. Five specimens were tested for each condition, as shown in Fig. 2(b). A total of 155 specimens were tested, and 90 specimens were subjected to sustained loads. The tested unconditioned ultimate tensile strength was  $1,398 \pm 30$  MPa, the tensile modulus was  $46 \pm 0.4$  GPa, and the ultimate strain was  $3.0 \pm 0.07\%$ .

### Test Setup and Procedure

A specific device was developed by the research group to apply the sustained load on the BFRP bars, as shown in Fig. 3(a). First, the axial load was applied using a hydraulic jack with the aid of a reaction frame and connector, and the applied load was monitored simultaneously using a load sensor. Second, the nut-2 was tightened with a wrench and the hydraulic jack was unloaded after



**Fig. 2.** Tensile specimens: (a) dimension of the specimens for tensile test (mm); (b) anchored BFRP bars (images by Gang Wu)



**Fig. 3.** Specimen preparation: (a) device for axial-load application; (b) installed PVC reservoir; (c) prepared stressed specimen (images by Gang Wu)

the load level was reached. According to preliminary research on the developed device, a sustained load could be effectively maintained by the nut during the experiment. The middle test section of the bar was inserted into a PVC tube measuring 20 mm in diameter and 240 mm in length, which served as a reservoir, as shown in Fig. 3(b). After the sustained load was applied, the two ends of the PVC tube were sealed with glass cement; the middle pipe tee, which was installed for the addition of the corrosive solution, was covered by a plastic film secured with rubber bands after perfusion. The prepared stressed specimen is shown in Fig. 3(c). The preparation of the unstressed specimens was the same as that of the stressed specimens except for the applied load. During the test, the pH value of the solutions was held constant according to regular inspection and solution supplement. Specimens were placed in a temperature-controlled cabinet ( $0 - 80 \pm 3^\circ\text{C}$ ) through the aging duration. A summary of the test program is presented in Table 2.

### Simulated Corrosive Solutions

Because of the limited volume of the reservoir constructed from the PVC tube, the simulated concrete-pore alkaline solution consisted of 2 g  $\text{Ca}(\text{OH})_2$ , 0.9 g NaOH, and 4.2 g KOH in 1 L deionized water based on ACI440.3R-04 (ACI 2004) with selected modifications, and the measured pH values were in the 12.9–13.1 range. The acid solution contained 1.58 g concentrated sulfuric acid with a mass fraction of 98.3% in 1 L deionized water, and the test pH value was 1.5. Following ASTM D1141-98 (ASTM 2008),

**Table 2.** Test Program

| Solution types  | Stress level | Temperature ( $^\circ\text{C}$ ) | Aging (days)   |    |    |
|-----------------|--------------|----------------------------------|----------------|----|----|
|                 |              |                                  | 21             | 42 | 63 |
| Alkaline        | 0            | 25                               | 5 <sup>a</sup> | 5  | 5  |
|                 | 0            | 40                               | 5              | 5  | 5  |
|                 | 0            | 55                               | 5              | 5  | 5  |
|                 | 20%          | 40                               | 5              | 5  | 5  |
|                 | 20%          | 55                               | 5              | 5  | 5  |
|                 | 40%          | 25                               | 5              | 5  | 5  |
|                 | 40%          | 40                               | 5              | 5  | 5  |
|                 | 60%          | 40                               | 5              | 5  | 5  |
| Deionized water | 0            | 40                               | /              | 5  | /  |
|                 | 10%          | 40                               | /              | 5  | /  |
| Salt            | 0            | 40                               | /              | 5  | /  |
|                 | 10%          | 40                               | /              | 5  | /  |
| Acid            | 0            | 40                               | /              | 5  | /  |
|                 | 10%          | 40                               | /              | 5  | /  |

<sup>a</sup>Number of specimens used for each group.

the composition of the salt solution was 24.53 g NaCl, 5.20 g  $\text{MgCl}_2$ , 4.09 g  $\text{Na}_2\text{SO}_4$ , and 1.16 g  $\text{CaCl}_2$  in 1 L deionized water.

### Scanning Electron Microscopy

Scanning electron microscopy observations and image analysis were performed to evaluate the microstructure and the integrity



of the stress-free BFRP samples in alkaline solution. The samples observed in SEM were reference specimens and stress-free specimens conditioned in 55°C alkaline solution for the longest duration, which was 63 days. The specimens were first embedded in epoxy resin, cut with a low-speed saw, and then polished with sandpaper and fluffy cloth. Microstructural observations were thereafter performed on a Carl Zeiss Ultra Plus Field Emission SEM (Oberkochen, Germany). The potential degradation of the matrix, basalt fibers, or interface, if any, was observed. These observations were mainly focused on the outer parts of the bar samples because they are the main portion subjected to and affected by aggressive exposures.

## Test Results and Discussion

For the bars coupled with sustained loads, after the specified exposure times were reached, the load force was unloaded and the PVC pipe was removed. The bar surface was washed with deionized water and dried in air for three days before testing. For the unstressed bars, the bar ends were anchored with steel tubes after washing and drying. The tensile property test was conducted using a universal MTS tension-compression test machine with a capacity of 500 kN. The loading rate was set at 2 mm/min up to failure. The results are shown in Table 3, which also lists the coefficients of variation. The test results showed that the variation of the elastic modulus was small and displayed no obvious regularity. The elastic modulus could thus be treated as unaffected and is not discussed further.

## Effect of Temperature

The diffusion rate of the corrosive media (e.g.,  $H_2O$ ,  $OH^-$ ) are accelerated by the elevated temperatures, which results in a larger affected cross-sectional area of the FRP bars. Fig. 4 compares the strength retention for stress-free bars in alkaline solution at different temperatures, and it can be observed that the higher the temperature, the faster the degradation. The degradation rate was slow at 25°C, and the strength retention was 94% even after 63 days of immersion. However, the strength retention was 90% at 40°C after only 21 days. The degradation rate was faster after the initial 21 days at 40°C; however, the rate subsequently slowed and the observed strength retention was 85% after 63 days of immersion. The degradation rate at 55°C was obviously faster than that at the previous two temperatures. The strength retention dropped to 82% after only 21 days and dropped further to 69% after 63 days. Additionally, because of the short duration, no slowing trend was observed.

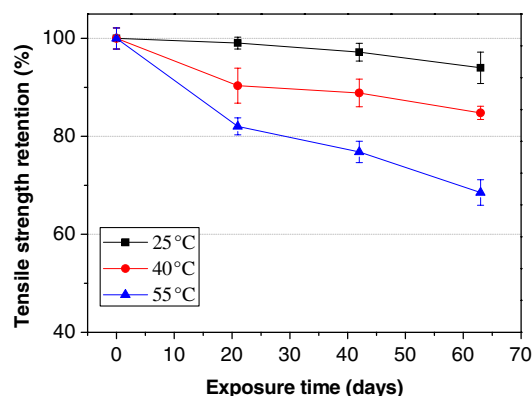
## Effect of Solution Type

Fig. 5 compares tensile-strength retention in the four types of solution after 42 days of immersion at 40°C. It can be observed that the alkaline-resistance property of the BFRP bars was not as good as the properties in deionized water, salt, and acid solutions. According to the tested data, after immersion in alkaline solution at 40°C for 42 days, the tensile strength of the BFRP bar was reduced by 6%, which was 6% and 4% more than that of the bars immersed in

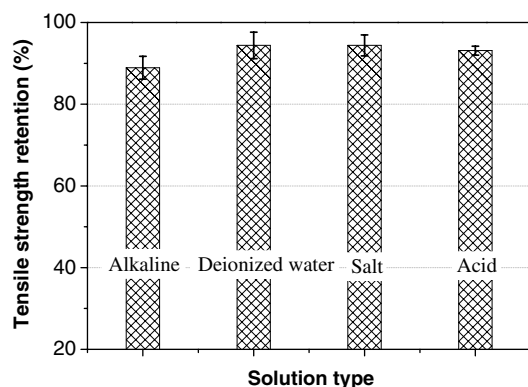
**Table 3.** Test Results

| Solution type   | Stress level (%)        | Temperature (°C) | Aging (days) | Tensile strength (MPa) |               |                     | Elastic modulus (GPa) |        |      |
|-----------------|-------------------------|------------------|--------------|------------------------|---------------|---------------------|-----------------------|--------|------|
|                 |                         |                  |              | Mean                   | Retention (%) | CV <sup>a</sup> (%) | Mean                  | CV (%) |      |
| Alkaline        | Unconditioned specimens |                  |              | 1,398                  | 100.0         | 2.16                | 46                    | 0.78   |      |
|                 | 0                       | 25               | 21           | 1,385                  | 99.1          | 1.20                | 45                    | 0.43   |      |
|                 |                         |                  | 42           | 1,359                  | 97.2          | 1.82                | 45                    | 0.27   |      |
|                 |                         |                  | 63           | 1,315                  | 94.0          | 3.21                | 45                    | 0.41   |      |
|                 |                         | 40               | 21           | 1,263                  | 90.3          | 3.57                | 45                    | 0.58   |      |
|                 |                         |                  | 42           | 1,243                  | 88.9          | 2.81                | 44                    | 0.40   |      |
|                 |                         |                  | 63           | 1,186                  | 84.8          | 1.34                | 44                    | 0.46   |      |
|                 |                         | 55               | 21           | 1,147                  | 82.0          | 1.73                | 46                    | 2.10   |      |
|                 |                         |                  | 42           | 1,074                  | 76.8          | 2.18                | 45                    | 1.20   |      |
|                 |                         |                  | 63           | 958                    | 68.5          | 2.60                | 45                    | 1.12   |      |
|                 | 20                      | 40               | 21           | 1,313                  | 93.9          | 0.98                | 45                    | 0.72   |      |
|                 |                         |                  | 42           | 1,271                  | 90.9          | 1.53                | 45                    | 0.61   |      |
|                 |                         |                  | 63           | 1,166                  | 83.4          | 5.01                | 45                    | 2.14   |      |
|                 |                         | 55               | 21           | 1,183                  | 84.6          | 5.44                | 45                    | 0.53   |      |
|                 |                         |                  | 42           | 974                    | 69.6          | 6.97                | 45                    | 3.54   |      |
|                 |                         |                  | 63           | 852                    | 60.9          | 6.24                | 45                    | 2.35   |      |
|                 |                         | 40               | 25           | 21                     | 1,291         | 92.3                | 1.60                  | 45     | 0.77 |
|                 |                         |                  |              | 42                     | 1,277         | 91.3                | 1.98                  | 45     | 1.30 |
|                 |                         |                  |              | 63                     | 1,229         | 87.9                | 1.61                  | 45     | 0.81 |
|                 | 40                      |                  | 21           | 1,227                  | 87.8          | 2.16                | 45                    | 0.39   |      |
|                 |                         |                  | 42           | 1,130                  | 80.8          | 3.41                | 44                    | 0.66   |      |
|                 |                         |                  | 63           | 1,092                  | 78.1          | 1.62                | 44                    | 1.64   |      |
|                 | 60                      | 40               | 21           | 1,190                  | 85.1          | 3.20                | 43                    | 2.12   |      |
|                 |                         |                  | 42           | Rupture                | /             | /                   | /                     | /      |      |
|                 |                         |                  | 63           | Rupture                | /             | /                   | /                     | /      |      |
| Deionized water | 0                       | 40               | 42           | 1,320                  | 94.4          | 3.23                | 45                    | 0.56   |      |
|                 |                         |                  | 42           | 1,275                  | 91.2          | 4.45                | 46                    | 0.36   |      |
| Salt            | 0                       | 40               | 42           | 1,320                  | 94.4          | 2.58                | 45                    | 0.21   |      |
|                 |                         |                  | 42           | 1,294                  | 92.6          | 2.59                | 45                    | 0.67   |      |
| Acid            | 0                       | 40               | 42           | 1,301                  | 93.1          | 1.09                | 45                    | 0.17   |      |
|                 |                         |                  | 42           | 1,291                  | 92.3          | 2.05                | 45                    | 0.32   |      |

<sup>a</sup>Coefficient of variation.



**Fig. 4.** Comparison of strength retention for stress-free bars in alkaline solution at different temperatures

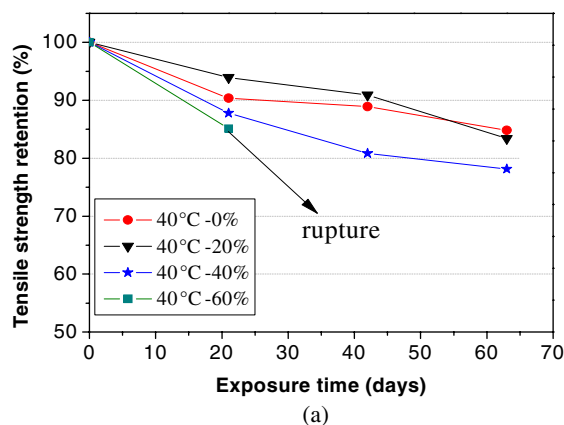


**Fig. 5.** Comparison of residual tensile strength in different types of solution

the deionized water, salt, and acid solutions, respectively, under the same conditions.

### Effect of Sustained Load

The effect of the coupled-stress levels on the tensile-strength retention was comparatively studied, and a summary of the test results is given in Table 3 and Fig. 6. In general, the coupled stresses increased the corrosion of the BFRP bar.



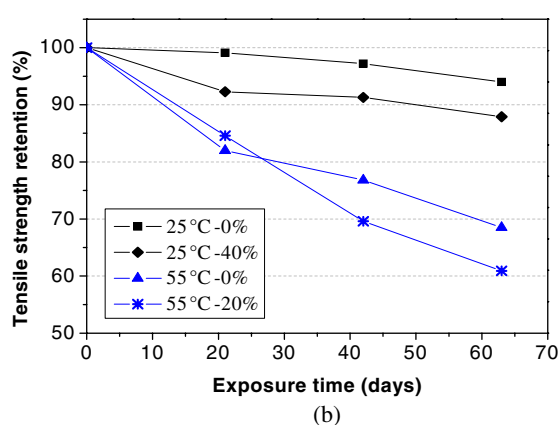
According to the curves in Fig. 6, there are important differences in the acceleration effects of different stress levels. Fig. 6(a) shows the data at 40°C in alkaline solution. The curves of the bars with stress levels of 0 and 20% were very close to each other and the reductions at a 20% stress level were even less than those at the 0% stress level at 21 and 42 days. However, when the stress level reached 40%, the degradation rate was obviously accelerated. For the high stress level of 60%, the bars ruptured after approximately 42 days. Fig. 6(b) shows the data at 25 and 55°C in alkaline solution. The coupled-stress level of 40% obviously accelerated the degradation rate at 25°C. For the two curves at 55°C with stress levels of 0 and 20%, the degradation at 21 days was similar, whereas the stressed bars showed greater reductions than did the unstressed bars at 42 and 63 days. This result may be due to the relatively more extensive strength degradation at 55°C, and the coupled effect of stress was obvious because of the increased effective stress level (applied sustained load divided by residual tensile strength).

An investigation of the coupled effect of a sustained load on the BFRP bars in de-ionized water, acid, and salt solution was also conducted. To verify whether a low coupled-stress level had a minor effect on the degradation rate, a stress level of 10% was chosen. According to the data in Table 3, the tensile strength of BFRP bars in deionized water and salt and acid solutions with a stress level of 10% was reduced by 3, 2, and 1% over that of the unstressed bars, respectively. These negligible differences further verified the conclusion that a low coupled-stress level has a minor effect on the degradation of the BFRP bars.

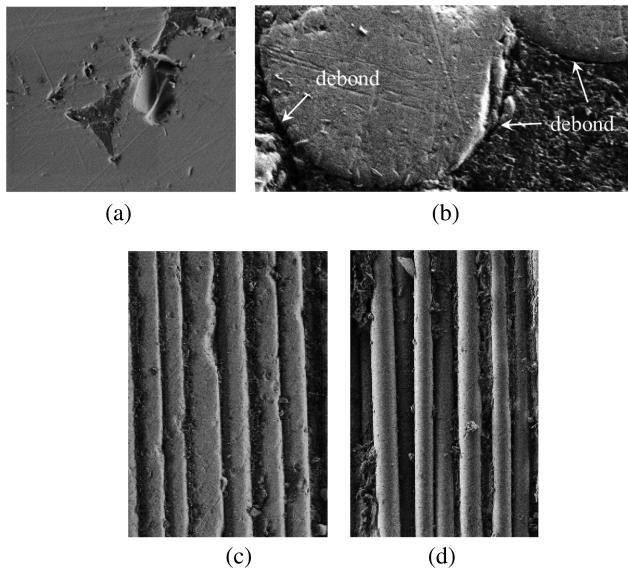
### Microstructural Analysis and Degradation Mechanism

The SEM images of BFRP bars before and after corrosion are shown in Fig. 7. There was obvious fiber/matrix-interface debonding in the regions close to the bar surface after corrosion. However, Fig. 7(d) shows that the surface of the exposed basalt fiber was smooth and that there was no visible damage. This indicated that the degradation of tensile strength was derived from the damage to the fiber/matrix interface rather than to the fibers themselves in the test cycle.

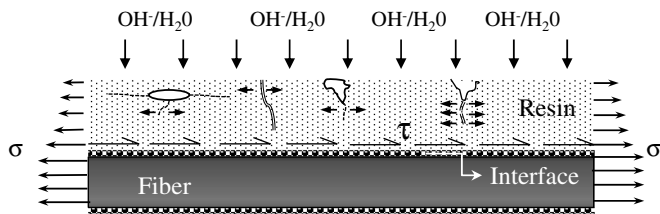
On the basis of the degradation mechanism revealed by SEM images and similar studies on GFRP bars (Benmokrane et al. 2002; Nkurunziza et al. 2005; Debaiky et al. 2006), the acceleration mechanism of stress on BFRP bars is shown in Fig. 8. The existing stresses in the resin matrix promoted the appearance and extension of microcracks, causing the corrosive media to be more likely to



**Fig. 6.** Effects of coupled sustained stress on the tensile-strength retention of BFRP bars immersed in alkaline solution: (a) effect of stress level at a constant temperature; (b) effect of stress level and temperature



**Fig. 7.** Micrographs of the fiber/matrix interface: (a) transversal section of reference sample; (b) transversal section of the stress-free sample conditioned in 55°C alkaline solution for 63 days; (c) longitudinal section of the reference sample; (d) longitudinal section of the stress-free sample conditioned in 55°C alkaline solution for 63 days (images by Gang Wu)



**Fig. 8.** Acceleration mechanism of stress

erode and damage the fiber/resin interface. Because the interface plays an important role in the load transfer from matrix to fiber, the tensile strength is reduced when the interface is destroyed.

According to Benmokrane et al. (2002) “for GFRP bars embedded in moist concrete under various sustained stress levels, three types of stress corrosion mechanisms have been identified: stress dominated, crack propagation dominated, and diffusion dominated.” According to the test data for the BFRP bars shown here in Fig. 6 under different stress levels in alkaline solution, it was thought that similar mechanisms exist in BFRP reinforcement. Low stress levels (perhaps less than 20%) do not expand the matrix voids enough to create cracks. No direct attack on the fiber/matrix interface takes place, and the surrounding medium penetrates the bar only by diffusion. This is the so-called diffusion-dominated stage. At moderate stress levels (e.g., 20–40%), the matrix suffers from micro-cracking, which allows the environment to attack the fiber/matrix interface. The degree of crack propagation is a critical factor for the residual tensile strength. Therefore, this phase was named the crack-propagation-dominated stage. At high stress levels, because of the creep-fracture characteristics of the BFRP bars, the bars fracture even without contact with corrosive solution after a period of time. Experimental tests indicated that the coupled corrosive solution (i.e., alkaline) shortens the time needed for the bar failure. That is why this phase is called the stress-dominated stage.

## Prediction of Long-Term Behavior in Alkaline Solution

### Arrhenius Relationship

In the Arrhenius relationship, the degradation rate is expressed by Eq. (1), as proposed by Nelson (1990) and as widely adopted by other researchers (Chen et al. 2006; Robert and Fam 2012; Davalos et al. 2012; Robert and Benmokrane 2013):

$$k = A \exp(-E_a/RT) \quad (1)$$

where  $k$  = degradation rate (1/time);  $A$  = constant of the material and degradation process;  $E_a$  = activation energy;  $R$  = universal gas constant; and  $T$  = Kelvin temperature. The primary assumption of predictions using the Arrhenius relationship is that the single dominant degradation mechanism does not change with time and temperature during the exposure, but the degradation rate is accelerated with the increase in temperature. Eq. (1) can be transformed into Eqs. (2) and (3):

$$\frac{1}{k} = \frac{1}{A} \exp(E_a/RT) \quad (2)$$

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R} \frac{1}{T} - \ln A \quad (3)$$

Eq. (2), shows that the degradation rate  $k$  can be expressed as the inverse of the time required for a material property to reach a given value. Eq. (3), shows that the logarithm of the time required for a material property to reach a given value is a linear function of  $1/T$  with a slope of  $E_a/R$ .

### Degradation Rule

Degradation data for at least three different temperatures are required for predictions using Arrhenius theory. Therefore, the described prediction procedure was conducted only on an unstressed bar in alkaline solution. Before the long-term prediction procedure is conducted, the rule for the degradation data must be established. Two types of commonly used degradation models exist:

$$Y = a \log(t) + b \quad (4)$$

$$Y = 100 \exp(-t/\tau) \quad (5)$$

where  $Y$  = tensile-strength retention (%);  $t$  = exposure time in days;  $a$  and  $b$  = regression constants; and  $\tau$  = fitted parameter.

The model of Eq. (4) was adopted by Caceres et al. (2000) and Deje (2001), and was also presented by Bank et al. (2003). However, as shown in Fig. 9(a), the Arrhenius plots fitted using the data here were not parallel to each other, which violated the fundamental assumption of this model. A similar problem was also reported by Gonenc (2001). The model of Eq. (5) was successfully adopted by Chen et al. (2006) and Davalos et al. (2012) in performance prediction research for GFRP bars. The degradation mechanism for this model was assumed to be debonding at the fiber/matrix interface, an assumption in line with the observed deterioration according to SEM images. Thus, the model of Eq. (5) was adopted in this study. On the basis of Eq. (5), the fitted curves are shown in Fig. 9(b) and the obtained parameters are listed in Table 4. The correlation coefficients ( $R^2$ ) of all of the curves are greater than 0.85.

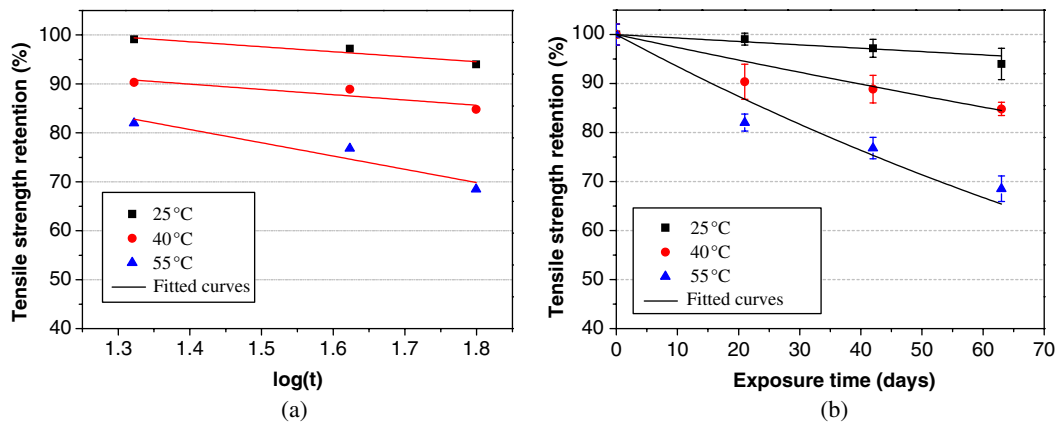


Fig. 9. Fitted curves for tensile-strength retention versus time: (a) curves of Eq. (4); (b) curves of Eq. (5)

Table 4. Coefficients of the Regression Equations in Eq. (5)

| Temperature (°C) | Stress level | BFRP bars in alkaline solution |       |
|------------------|--------------|--------------------------------|-------|
|                  |              | $\tau$                         | $R^2$ |
| 25               | 0            | 1,412.8                        | 0.85  |
| 40               | 0            | 374.4                          | 0.96  |
| 55               | 0            | 148.4                          | 0.91  |

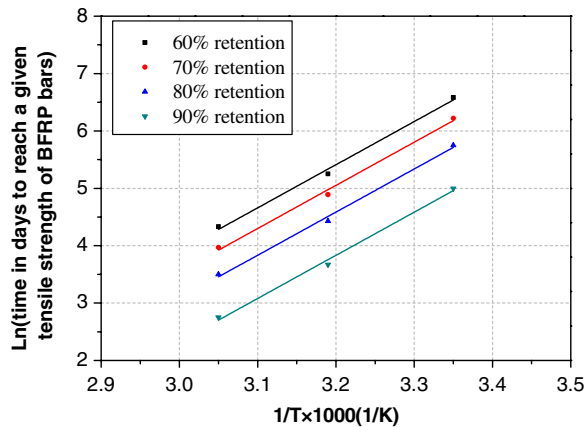


Fig. 10. Arrhenius plots of tensile-strength retention

#### Prediction Procedure for an Unstressed BFRP Bar

As shown in Fig. 10, with the regression coefficients listed in Table 4 the times needed for the tensile-strength retention to reach 60, 70, 80, and 90% at 25, 40, and 55°C were obtained from Eq. (5). Next, Eq. (3) was fitted to those values, and the straight lines were nearly parallel. The slopes of the straight lines ( $E_a/R$ ) and the correlation coefficients ( $R^2$ ) are listed in Table 5. The  $R^2$  values were

Table 5. Coefficients of the Regression Equations in Eq. (3)

| Tensile-strength retention (%) | BFRP bars in alkaline solution |       |
|--------------------------------|--------------------------------|-------|
|                                | $E_a/R$                        | $R^2$ |
| 60                             | 7,519                          | 0.99  |
| 70                             | 7,519                          | 0.99  |
| 80                             | 7,518                          | 0.99  |
| 90                             | 7,519                          | 0.99  |

all greater than 0.99, and the obtained  $E_a/R$  values were approximately 7,519. In a similar study (Chen et al. 2006), two  $E_a/R$  values of 1,420 and 4,890 were obtained for two types of 9.53-mm GFRP bars with different types of E-glass fibers, respectively. DeJke (2001) obtained an  $E_a/R$  value of 6,300 for GFRP bars embedded in moist concrete. Thus, if the  $E_a/R$  value is obtained for a specific FRP bar type and surrounding condition, it may not be applicable to other types of FRP bars used in different environments. From the literature review, it was found that numerous existing studies have been conducted on GFRP bars but their results have rarely been replicated on BFRP bars. This paper fills this research gap.

According to Eq. (1), the time-shift factor (TSF) for the tensile strength to reach the same value (represented by  $c$ ) at temperatures  $T_1$  and  $T_0$  can be obtained from the previous Arrhenius plots. The TSF can be expressed as

$$\text{TSF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{A \exp(-E_a/RT_1)}{A \exp(-E_a/RT_0)} = \exp \left[ \frac{E_a}{R} \left( \frac{1}{T_0} - \frac{1}{T_1} \right) \right] \quad (6)$$

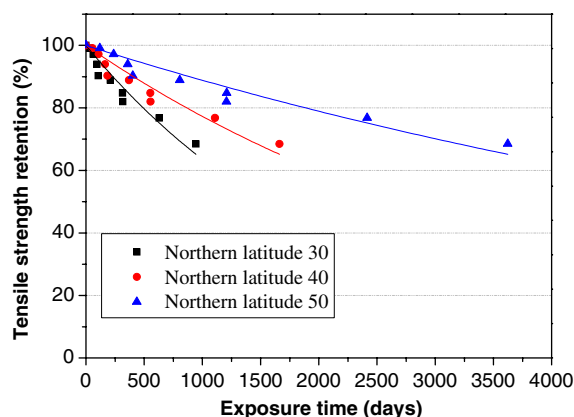
The TSF with reference temperatures  $T_0$  equal to 20.3, 14.0, and 5.7°C, representing the annual temperature at northern latitudes 30°, 40°, and 50°, respectively, are listed in Table 6.

In the northern latitude 50° area, for example, the tensile-strength retention data in Fig. 4 were transformed into those of Fig. 11 by multiplying the exposure times at 55, 40, and 25°C with the corresponding TSF values. The master curve versus aging time at the annual temperature of the northern area at latitude 50° could be obtained by fitting Eq. (5) to the data in Fig. 11. Following these steps, the master curves for northern areas at latitudes 30° and 40° were obtained, also as shown in Fig. 11. As listed in Table 7, all of the fitted curves had correlation coefficients of 0.92, confirming the validity of this procedure. Finally, using the fitted parameter  $\tau$  and

Table 6. Time-Shift Factor

| Temperature (°C) | Time-shift factor (TSF)        |                                |                               |
|------------------|--------------------------------|--------------------------------|-------------------------------|
|                  | (Northern latitude 30°) 20.3°C | (Northern latitude 40°) 14.0°C | (Northern latitude 50°) 5.7°C |
| 55               | 15.0                           | 26.4                           | 57.5                          |
| 40               | 5.0                            | 8.8                            | 19.2                          |
| 25               | 1.5                            | 2.6                            | 5.7                           |





**Fig. 11.** Master curves of BFRP bars exposed to different annual temperatures

**Table 7.** Coefficients of the Regression Equations for the Master Curves

| Northern latitude (°) | Mean annual temperature (°C) | BFRP bars in alkaline solution |       |
|-----------------------|------------------------------|--------------------------------|-------|
|                       |                              | $\tau$                         | $R^2$ |
| 30                    | 20.3                         | 2,206.5                        | 0.92  |
| 40                    | 14.0                         | 3,882.1                        | 0.92  |
| 50                    | 5.7                          | 8,457.5                        | 0.92  |

Eq. (5), the exposure times required for any tensile-strength retention could be easily obtained.

From the master curves, the tensile-strength retention of a BFRP bar in alkaline solution dropped to 50% after 4.2 years, 7.4 years, and 16.1 years at 20.3, 14.0, and 5.7°C, which represent the annual temperatures of the northern areas at latitudes 30°, 40°, and 50°, respectively. The predicted service life for a specific GFRP reinforcement directly exposed to alkaline solution was between 1.6 and 4.6 years, as reported by Sen et al. (2002). According to the predictions made by Chen et al. (2006) on 9.5-mm GFRP bars, the exposure times required for a tensile-strength retention of 50% were 0.5–3 years at 20°C in alkaline solution. The adopted BFRP bars were directly immersed in alkaline solution, and thus this conclusion should be considered conservative. In fact, concrete-pore solution in ambient air conditions was definitely less aggressive than alkaline solution. In a field investigation of a GFRP bar by Mufti et al. (2005), no degradation was observed after service for 6 to 8 years. Thus, long-term data should be collected in field applications and the corresponding relationships between the accelerated corrosion in the laboratory and that in the real service environment must be established in future studies.

## Conclusion

This research was a component of an ongoing durability research program aimed at improving the durability performance of concrete structures in extreme corrosive environments using fiber-reinforced polymer (FRP) reinforcement bars. Based on the results of this research, the following conclusions may be drawn only for the newly developed 6-mm BFRP bars tested in this study:

1. The elastic modulus of the BFRP bars was quite stable and nearly unaffected by all four types of solutions and sustained stress levels applied.
2. The degradation rate was obviously accelerated by elevated temperature. For example, after conditioning in alkaline

solution for 63 days, the tensile-strength reductions for an unstressed BFRP bar were 31.5 and 6.0% at 55 and 25°C, respectively.

3. After immersion in 40°C alkaline solution for 42 days, the tensile strength of the BFRP bar was reduced by 5.5, 5.5, and 4.2% over bars immersed in deionized water, salt, and acid solutions, respectively, under the same conditions. Alkaline solution was the most aggressive condition for bare BFRP bars.
4. When the sustained-load level was less than 20% of the ultimate tensile strength, the coupled stress during corrosion had minor and negligible effects on the degradation process. When the stress level reached 40% of the ultimate strength, the degradation process was obviously accelerated. For a coupled-stress level of 60%, the failure of the bars was dominated by creep fracture, and the surrounding alkaline liquid media significantly accelerated the fracture process.
5. According to the long-term predictions based on Arrhenius theory, for the unstressed BFRP bar immersed in simulated alkaline solution, the times required for a tensile-strength reduction of 50% at 20.3, 14.0, and 5.7°C, which represent the annual temperatures of areas with northern latitudes 30°, 40°, and 50°, respectively, were 4.2 years, 7.4 years, and 16.1 years. Because the thin 6-mm and bare BFRP bars used in this paper were directly exposed to the solutions, the tested and predicted results should be considered conservative. Long-term data from samples exposed to outdoor environments are highly valuable for validating the data obtained in the laboratory.

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